

3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB

```
clc; clear; close all;

% Specifications
Fs=2000; Rp=1; Rs=40;
Wp=[300 600]/(Fs/2);
Ws=[200 700]/(Fs/2);

% Butterworth Filter
[n1,W1]=buttord(Wp,Ws,Rp,Rs);
[b1,a1]=butter(n1,W1,'bandpass');

% Chebyshev Type-I Filter
[n2,W2]=cheb1ord(Wp,Ws,Rp,Rs);
[b2,a2]=cheby1(n2,Rp,W2,'bandpass');

% Magnitude Response
figure;
freqz(b1,a1); title('Butterworth Filter');
figure;
freqz(b2,a2); title('Chebyshev Type-I Filter');

% Group Delay
figure;
grpdelay(b1,a1); hold on;
grpdelay(b2,a2);
legend('Butterworth','Chebyshev-I');

% Pole-Zero Plot
figure;
subplot(121),zplane(b1,a1),title('Butterworth')
subplot(122),zplane(b2,a2),title('Chebyshev-I')

% Impulse Response
figure;
```

```
subplot(211),impz(b1,a1),title('Butterworth Impulse')
```

```
subplot(212),impz(b2,a2),title('Chebyshev Impulse')
```

```
% Pole Magnitude
```

```
figure;
```

```
subplot(121),stem(abs(roots(a1)),'filled')
```

```
title('Butterworth Pole Magnitude')
```

```
subplot(122),stem(abs(roots(a2)),'filled')
```

```
title('Chebyshev Pole Magnitude')
```

```
disp(['Butterworth Order = ',num2str(n1)])
```

```
disp(['Chebyshev Order = ',num2str(n2)])
```







